रा.इ.सू.प्रो.सं
NIELIT
GUWAHATI

National Institute of Electronics and Information Technology
Ministry of Electronics and Information Technology
Government of India

DAY 1

R & RStudio Fundamentals

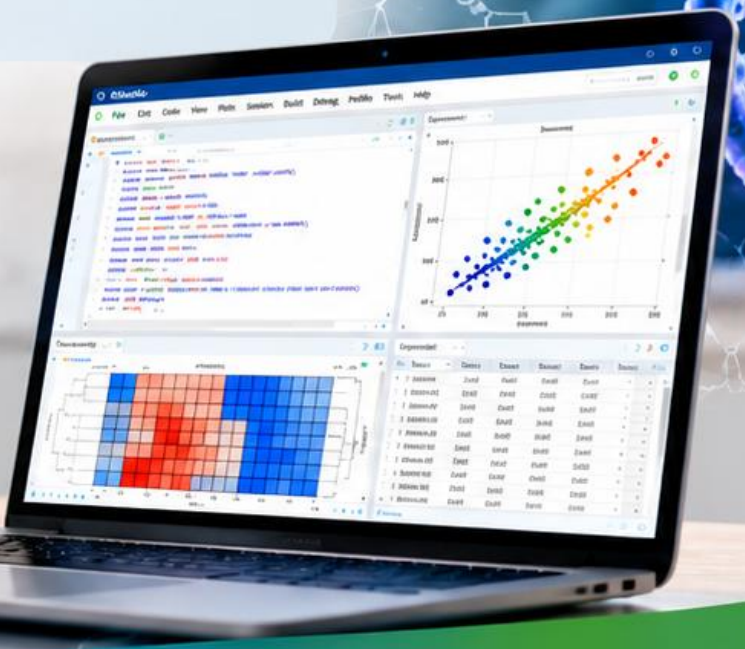
From the History of R to Basic R Programming for Biological Data Science

Duration:
1.5 Hours

Level:
Beginner

Target Audience:
Biology, Botany, Biotechnology, Bioinformatics and Life Science students/researchers





By
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1 R & RStudio Fundamentals

From the History of R to Basic R Programming



What is R?

R is an open-source programming language and software environment primarily designed for **statistical computing, data analysis** and **visualization**. It has become an important tool in biological research because modern biology generates large amounts of quantitative data.

R + RStudio®

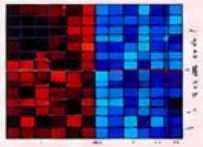
Where is R Used in Biology?



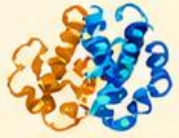
Plant Biology
Growth, physiology and stress experiments



Genomics
Genome-wide data and genetic variation



Gene Expression
RNA-Seq and transcriptomics



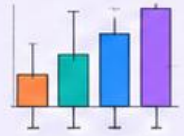
Proteomics
Protein abundance and expression analysis



Ecology
Species distribution and environmental observations



Phenotyping
Morphological and physiological measurements



Biostatistics
Hypothesis testing, correlation and statistical modelling

Example: Plant Height under Control and Drought Conditions

1 Biological Experiment

Measure height of 10 rice plants under Control and Drought conditions.



Control (Well-watered)



Drought (Water stress)

2 Enter Data in R

Create a data table with plant height measurements.

Plant	Control (cm)	Drought (cm)
1	14.2	8.1
2	15.1	7.5
3	13.8	9.0
4	16.0	8.4
5	14.5	7.8
6	15.3	8.1
7	13.9	7.9
8	14.8	8.3
9	15.5	7.6
10	14.7	8.0

3 Analyze Data in R

Calculate mean, standard deviation and compare groups.

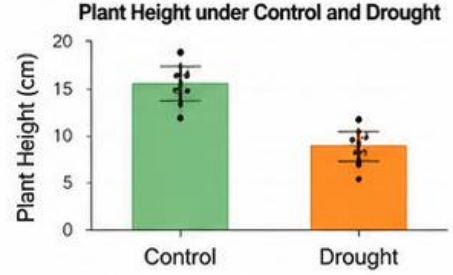
```
# Enter the data
control <- c(14.2, 15.1, 13.8, 16.0,
    14.5,15.3, 13.9, 14.8,15.5, 14.7)
drought <- c(8.1, 7.5, 9.0, 8.4, 7.8,
         8.1, 7.9, 8.3,7.6, 8.0)

# Calculate mean and SD
mean(control); sd(control)
mean(drought); sd(drought)

# Statistical test
t.test(control, drought)
```

4 Visualize the Results

Create a bar plot with error bars.



5 Interpret the Results

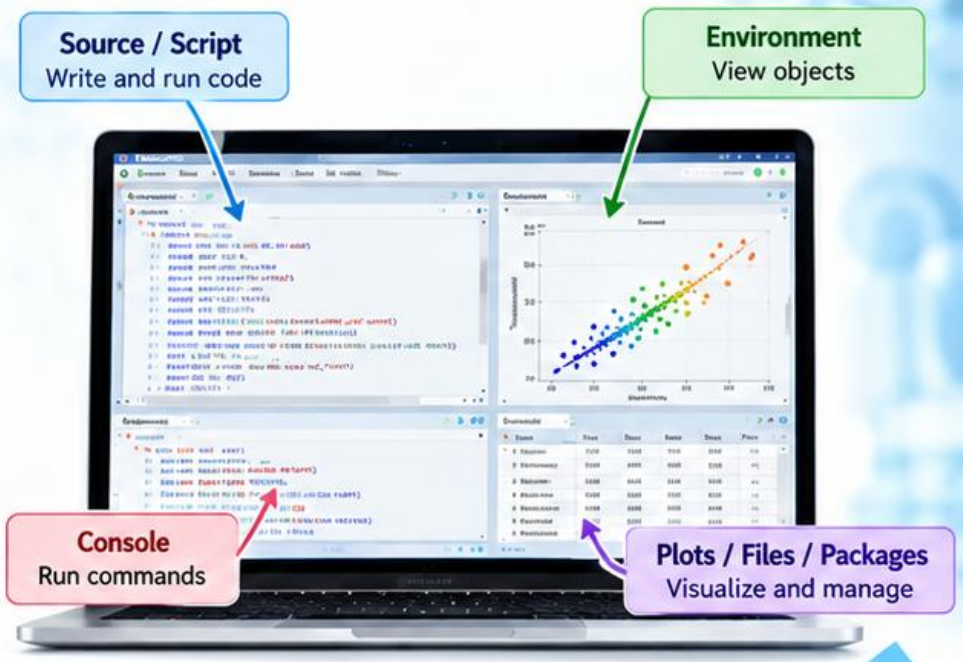
- Mean plant height is higher in control plants.
- Drought significantly reduces plant height ($p < 0.05$).
- R provides a fast, accurate and reproducible way to analyze and visualize biological data.

2 Learning Objectives

By the end of this session, participants should be able to:

1. Explain the **history and development of R.**
2. Understand the difference between **R and RStudio.**
3. Navigate the **RStudio interface.**
4. Write **basic R commands.**
5. Create and manipulate **R objects.**
6. Understand major **data types.**
7. Create **vectors, factors, matrices and data frames.**
8. Use basic **R functions.**
9. Handle simple **biological datasets.**
10. Develop a foundation for **advanced biological data analysis.**









3 Why Do Biologists Need R?

From Biological Data to Biological Insights



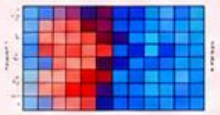
Modern Biology is Data-Driven

Modern biological research generates large, complex and multi-dimensional datasets. Traditional manual analysis or spreadsheets become difficult, time-consuming and error-prone when datasets are large.

A Typical Biological Experiment Generates



Hundreds of phenotype measurements
- Plant height
- Biomass
- Chlorophyll
- RWC, etc.



Thousands of genes
- RNA-Seq
- Transcriptomics
- Genomics
- Proteomics



Millions of sequencing reads
- RNA-Seq
- WGS
- Metagenomics



Multiple treatments and conditions
- Control vs Stress
- Time series
- Different tissues



Several biological replicates
- Biological
- Technical
- Environmental



Large ecological datasets
- Species data
- Environmental data
- Geographic data

Why Not Use Only Excel or Spreadsheets?

Spreadsheets are useful for small datasets, but **have limitations:**

- Difficult to handle large datasets
- Time-consuming manual calculations
- Higher risk of errors
- Limited statistical and visualization options
- Hard to reproduce the analysis

R provides a more systematic and powerful approach:

- Handles large and complex data
- Automates analysis using scripts
- Wide range of statistical methods
- High-quality visualizations
- Reproducible and shareable results

Example: Plant Height under Control and Drought Conditions

1 Input Data in R

```
# Plant height data (cm)
Control <- c(25, 27, 26, 28, 24)
Drought <- c(18, 20, 19, 21, 17)
```

Plant	Control (cm)	Drought (cm)
1	25	18
2	27	20
3	26	19
4	28	21
5	24	17

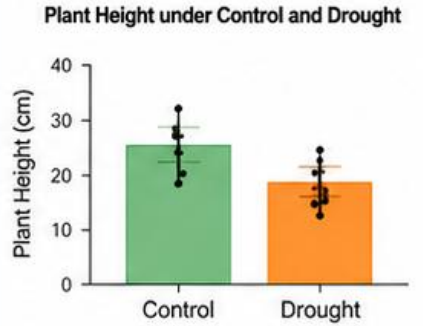
2 Calculate Statistics

```
mean(Control)
mean(Drought)
sd(Control)
sd(Drought)
```

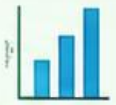
Output in R

```
> mean(Control)
[1] 26
> mean(Drought)
[1] 19
> sd(Control)
[1] 1.581139
> sd(Drought)
[1] 1.581139
```

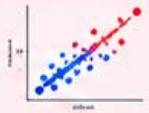
3 Visualize and Compare



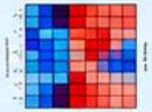
What Else Can R Do for Biologists?



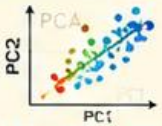
Statistical analysis
Mean, SD, hypothesis testing, ANOVA, etc.



Data visualization
High-quality graphs and publication figures



Gene expression analysis
RNA-Seq analysis, differential expression, heatmaps, PCA



Multivariate analysis
Correlation, regression, clustering



Reproducible research
Use scripts for automation and sharing



Wide range of biological applications
Genomics, proteomics, ecology, plant science, systems biology, etc.

Key Concept

"R is not just a calculator — it is a reproducible environment for transforming biological data into statistical evidence and visual insights."

Biological Question → Data → R Analysis → Visualization → Interpretation → Scientific Conclusion

4 A Short History of R

Origin: S Programming Language

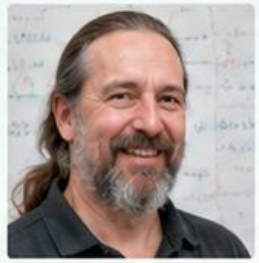
- R has its roots in the **S programming language**.
- The S language was developed at **Bell Laboratories (AT&T)** in the 1970s and 1980s for statistical computing and data analysis.
- S introduced many concepts that later influenced R, such as data objects, statistical modelling and an interactive environment.



Bell Laboratories
(S programming language)

Development of R

- R was developed by **Ross Ihaka** and **Robert Gentleman** at the **University of Auckland, New Zealand**.
- The first public release of R was in **1993**.
- The name R reflects the first names of its developers and its close relationship to the S language.
- R was designed as a free and open-source implementation of S, with improvements and new features.
- R quickly gained popularity in the statistical community due to its flexibility, extensibility and active user community.



Ross Ihaka
University of Auckland, New Zealand



Robert Gentleman
University of Auckland, New Zealand

THE UNIVERSITY OF AUCKLAND
NEW ZEALAND

1970s
Development of **S language** at Bell Laboratories for statistical computing.

1980s
S becomes popular in academia and industry for statistical analysis.

1991
Work on **R** begins by **Ross Ihaka** and **Robert Gentleman** at the **University of Auckland**.

1993
First public release of R (version 0.63).

2000
R 1.0.0 released.

2010
R becomes widely used across disciplines including **biology, genomics, ecology** and **bioinformatics.**

2020+
R continues to evolve with a large global community, thousands of packages and extensive applications in **data science, AI and bioinformatics.**

Important Idea

- R was not originally designed specifically for biology.
- Its powerful statistical and programming capabilities made it a highly adaptable tool for biological research.
- Today, R is widely used in many areas of biology, including transcriptomics, genomics, proteomics, systems biology, ecology, plant science and biomedical research.

S Language
(Bell Laboratories)
1970s–1980s

R
(University of Auckland)
1993–Present

Why R Became Popular?

- Free and open-source
- Powerful statistical capabilities
- Extensible through packages
- Active global community
- Cross-disciplinary applications (including biological research)

5 From S to R to Modern Biological Data Science

A journey from statistical computing to transformative biological research



S (1970s–1980s)

The foundation for statistical computing

- Developed at Bell Laboratories (AT&T).
- Provided a programming environment for statistics.
- Introduced key concepts such as data objects, statistical modelling and interactive analysis.



R (1990s–present)

An open-source evolution of S

- Developed by **Ross Ihaka** and **Robert Gentleman** at the University of Auckland, New Zealand.
- Open-source statistical programming environment.
- Extended and improved many ideas from S.
- Flexible, powerful and widely adopted across different fields.



Ross Ihaka
University of Auckland



Robert Gentleman
University of Auckland

CRAN (Comprehensive R Archive Network)

The global repository of R packages

- Central repository of thousands of user-contributed packages.
- Provides tools for statistical computing, data analysis and visualization.
- Regular updates and wide range of applications.
- Built and maintained by the global R community.



Bioconductor (2000s–present)

A specialized ecosystem for biological and genomic analysis

- Provides high-quality packages for genomics, transcriptomics, proteomics and systems biology.
- Developed and maintained by an active community of biologists, statisticians and data scientists.
- Widely used for the analysis of high-throughput biological data.

Bioconductor

Genomics · Transcriptomics · Proteomics

Pathway analysis · Variant analysis · Time-series analysis

Modern R Ecosystem (Present and Beyond)

R as a complete platform for modern biological data science

- **Statistics** — Hypothesis testing, modelling, machine learning
- **Data visualization** — High-quality and customizable graphics
- **Machine learning** — Classification, prediction, AI/ML approaches
- **Transcriptomics** — RNA-Seq, differential expression, network analysis
- **Genomics** — Genome analysis, variant calling, comparative genomics
- **Proteomics** — Protein expression and functional analysis
- **Epidemiology** — Disease analysis and population studies
- **Ecology** — Species distribution and environmental analysis

Key Message

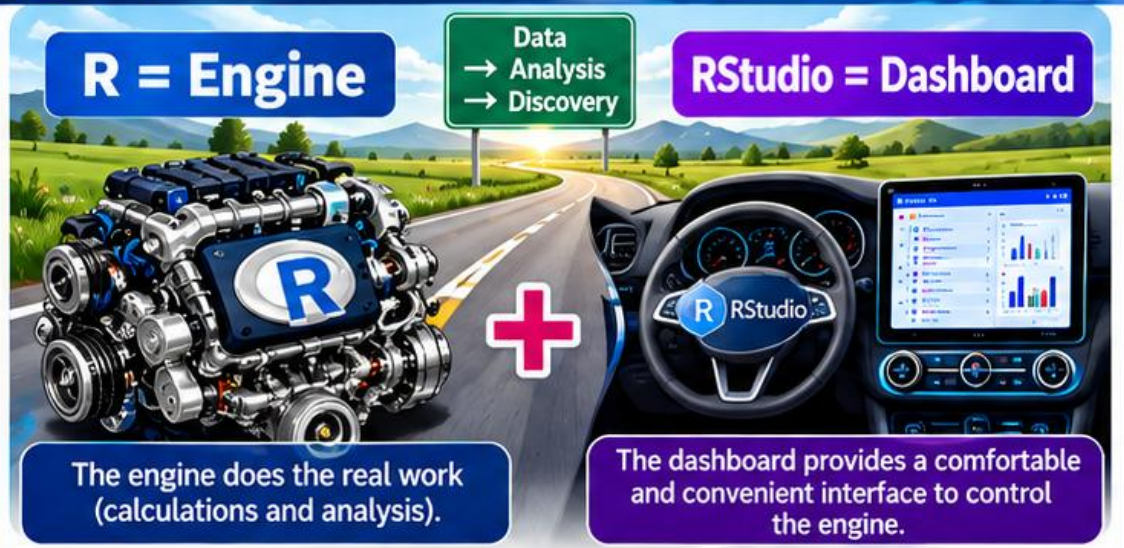
From statistical computing (S) to a powerful, open-source ecosystem (R), and specialized biological tools (Bioconductor), R has evolved into a **central platform for modern biological data science.**

6 R vs RStudio

Is R the same as RStudio? No.

R
The Programming Language
(Engine)

- Actual programming language and computational environment
- Designed for statistical computing, data analysis and visualization
- Executes commands and performs computations
- Can be used through different interfaces (R console, terminal, RStudio)
- Open-source and cross-platform (Windows, macOS, Linux)



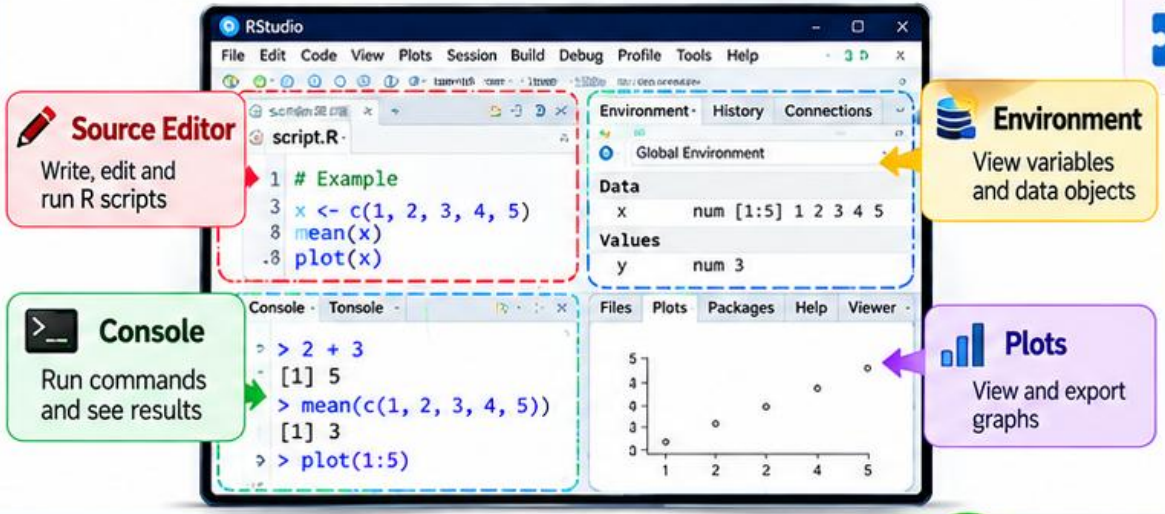
RStudio
The Integrated Development Environment (IDE)
(Dashboard)

- User-friendly interface to write, run and organize R code
- Helps manage files, packages, plots and data
- Integrated tools for debugging and project management
- Makes it easier and more efficient to work with R, especially for beginners
- Does not replace R — it uses the installed R version

Example: Simple Calculation

R Command (typed in R or RStudio)	R Output (in Console)
`2 + 3`	`[1] 5`

R performs the calculation.
RStudio provides a convenient place to write, run and organize this command.



Other Useful Panes in RStudio

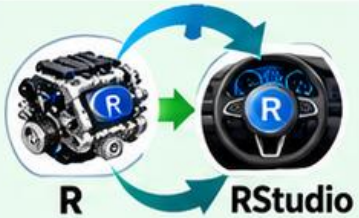
- **Files** — Navigate and manage project files
- **Packages** — Install and load R packages
- **Help** — Access documentation
- **Projects** — Organize complete analyses

Key Differences

Feature	R	RStudio
Type	Programming language	Integrated Development Environment (IDE)
Main function	Performs computations	Interface to work with R
Can it run R code?	Yes	Yes
Replaces R?	No	No (uses R)

Key Takeaway

"R is the engine.
RStudio is the dashboard.
Both work together."



7 Installing R and RStudio

Two components, one powerful setup for data analysis in biology!

1 Two components are normally installed:

R
The programming language

- Open-source
- Performs computations and analysis
- Can run from command line or within RStudio

RStudio
Graphical working environment (IDE)

- Provides a user-friendly interface
- Write, run and organize R code
- Manage files, packages, plots and data
- Includes tools for debugging and project management

2 Download and Install

A Install R

1. Go to https://cran.r-project.org
2. Click on your operating system (Windows / macOS / Linux)
3. Download the latest version
4. Run the installer and follow the instructions



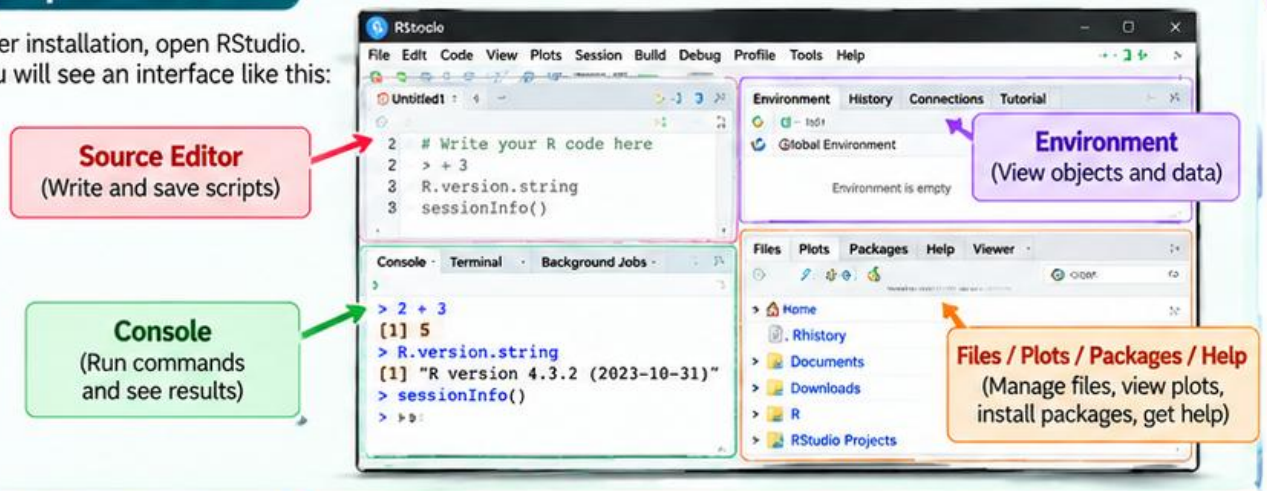
B Install RStudio

1. Go to https://posit.co/download/rstudio/
2. Choose the free version (RStudio Desktop)
3. Download for your operating system
4. Run the installer and follow the instructions



3 Open RStudio

After installation, open RStudio. You will see an interface like this:



4 Check the Installation

In the RStudio Console, type:

```
R.version.string
```

R will display the installed version.

```
> R.version.string
[1] "R version 4.3.2 (2023-10-31)"
```

Another useful command is:

```
sessionInfo()
```

This provides detailed information about your R environment.

5 Example Output of sessionInfo()

```
> sessionInfo()
R version 4.3.2 (2023-10-31)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64

Matrix products: default

locale:
  [1] LC_COLLATE=English_India.utf8
  [2] LC_CTYPE=English_India.utf8
  [3] LC_MONETARY=English_India.utf8
  [4] LC_NUMERIC=C
  [5] LC_TIME=English_India.utf8

attached base packages:
 [1] stats     graphics  grDevices utils
 [5] datasets  methods   base
```

Key Takeaway

- Install R (language)
- Install RStudio (IDE)
- Open RStudio
- Check version with `R.version.string`
- Get environment details with `sessionInfo()`